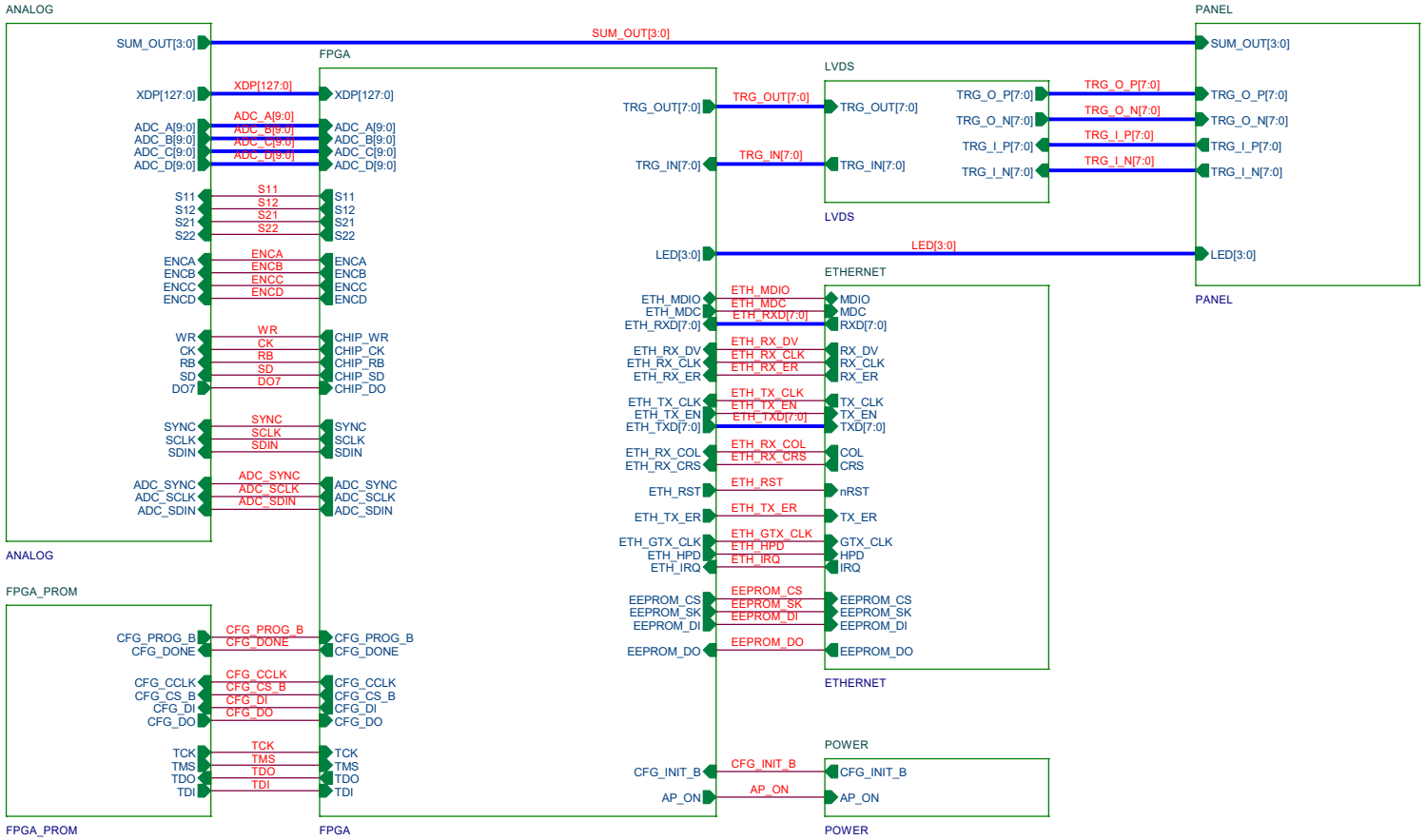
【Rev1.0】
FE2009bal μ-Gigabit_RO V1からの更新箇所
・FPGAをXC6SLX100-2FGG676CからXC7S100-2FGGA676Cに更新
・電源のシーケンシングに伴い、3.3Vにロードスイッチ(TPS22954DSQR)を追加
・電源のシーケンシングに伴いCFG_INIT_Bを電源のPGに接続
・旧コア電源用1.2V電源を削除
・FPGAのVCCINIT用1.0V電源にTPS7A9201DSKTを採用
・FPGAのVCCAUX用1.8V電源にTPS7A9201DSKTを採用
・アナログ電源を±2.5VをLDOで作成するためにLP38502ASD-ADJ/NOPBとLT3015EDD#PBFを採用
・アナログ電源のON/OFFを可能とするためにAP_ON信号を追加
・FPGAのコンフィグROMをXCF32PFSG48CからSPI-Flashに変更
・DIPスイッチを4回路から8回路に変更
・LVDSを汎用7出力,汎用7入力,1リセット入力から汎用8出力,汎用8入力に変更
・FPGA_RST廃止に伴いFPGAリコンフィグ用トランジスタ削除
・DACをAD5640BRJZ-1500RL7からAD5640CRMZ-1に変更
・ADCバイアス調整用VRをDACに変更。
・ネット名を判別しやすいようにトップシートに信号名追加

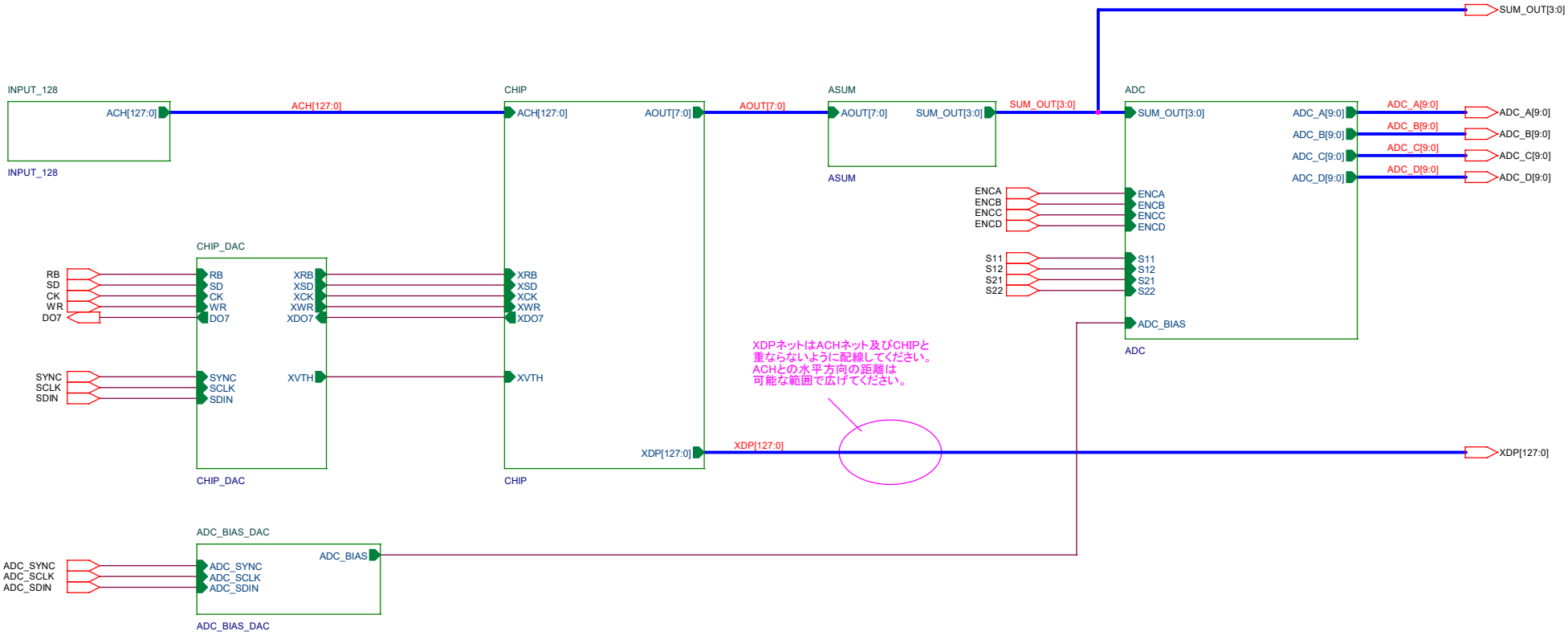
【Rev1.1】
基板パタン設計に伴うピンスワップを反映

【Rev1.2】
コメント修正のみ
・ET_TXの等長グループ(【等長配線A】)にETH_GTX_CLKを基準クロックとして追加

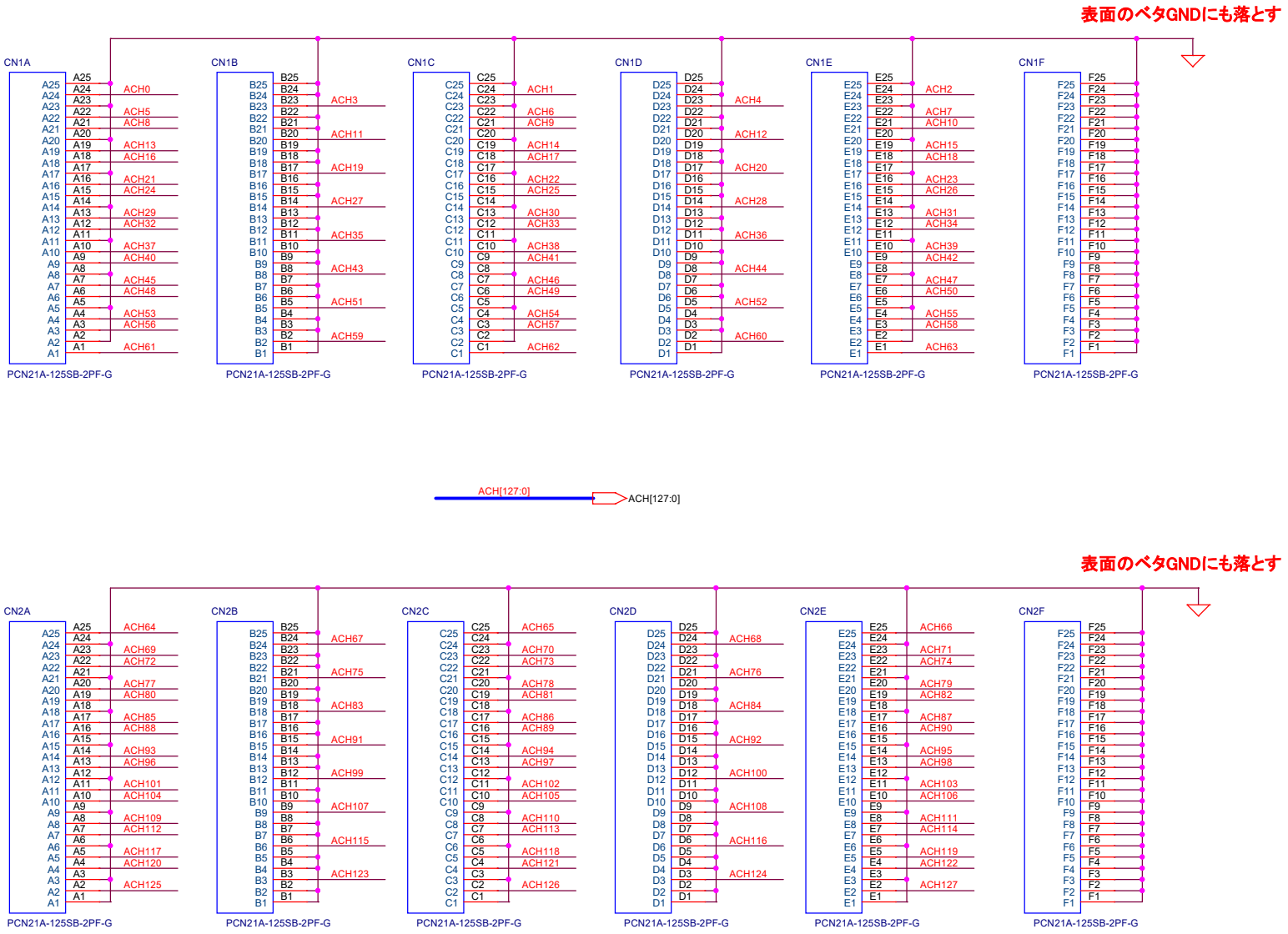
【Rev2.0】
FE2009bal μ-Gigabit_RO_V3からの変更
・基板名を「FE2009bal μ-Gigabit_RO V3.1」に変更
・スワップの指定を削除
・XDPネットの配線注意事項を追加
・TRIG_IN0のFPGA接続先をU31.U24からU31.T22へ変更



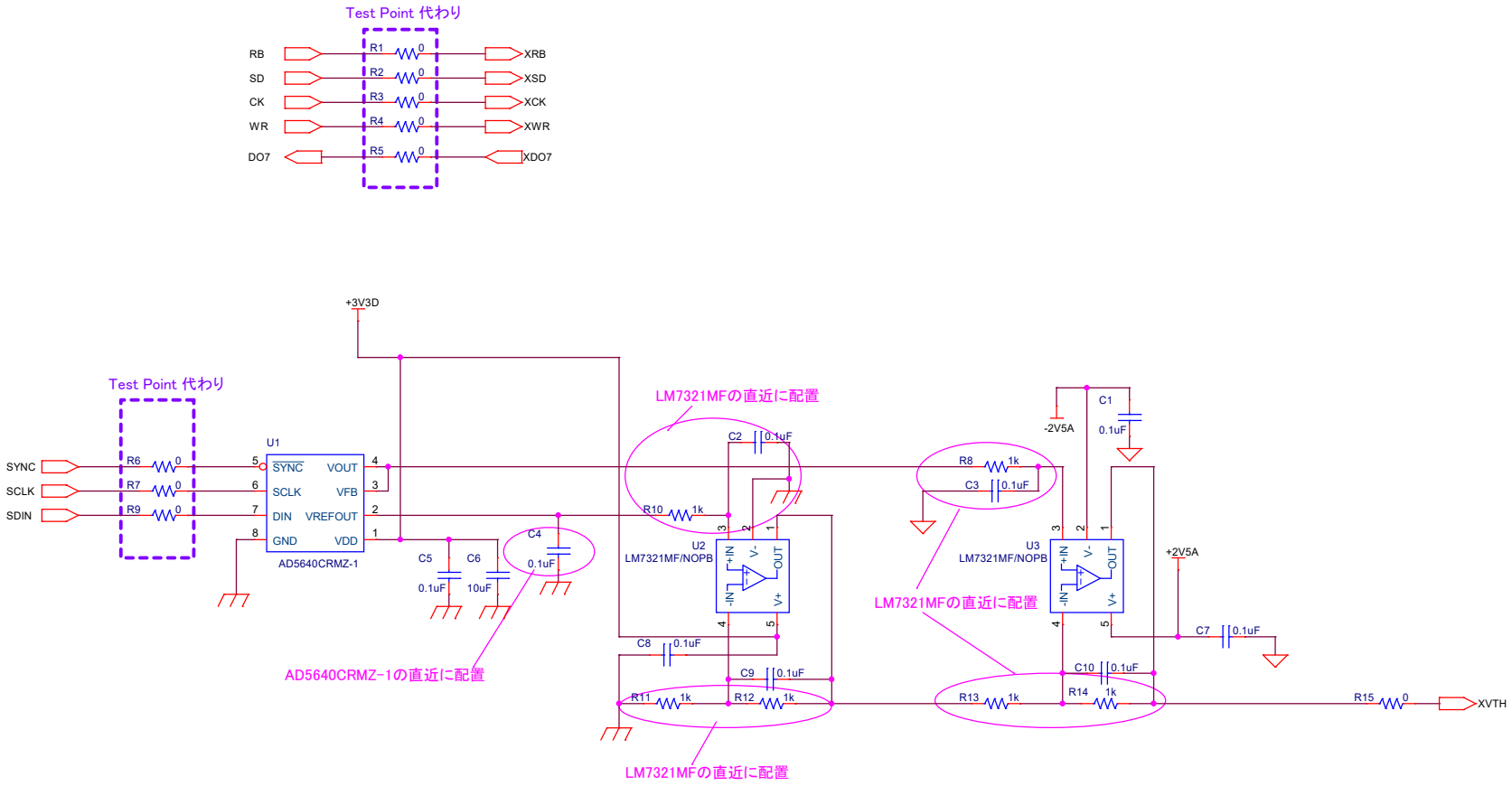
FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.	
Title: TOP			
Size: A3	Document Number: E2404001-001		Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 1 of 25		



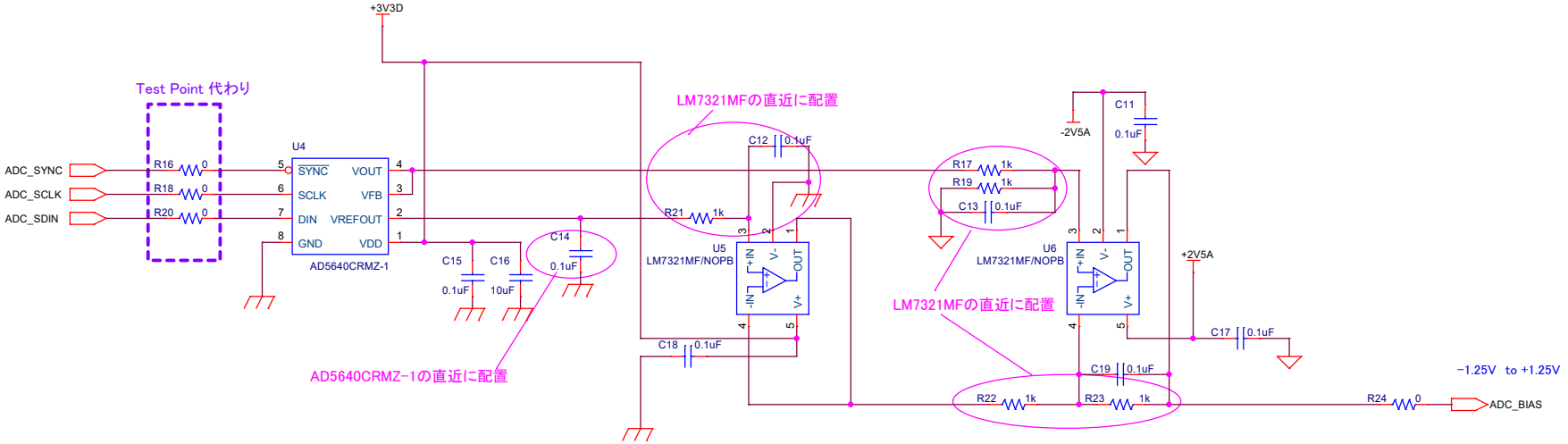
FE2009bal μ-Gigabit_RO V3.1	Bee Beans Technologies Co.,Ltd.	
Title: ANALOG		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 2 of 25	



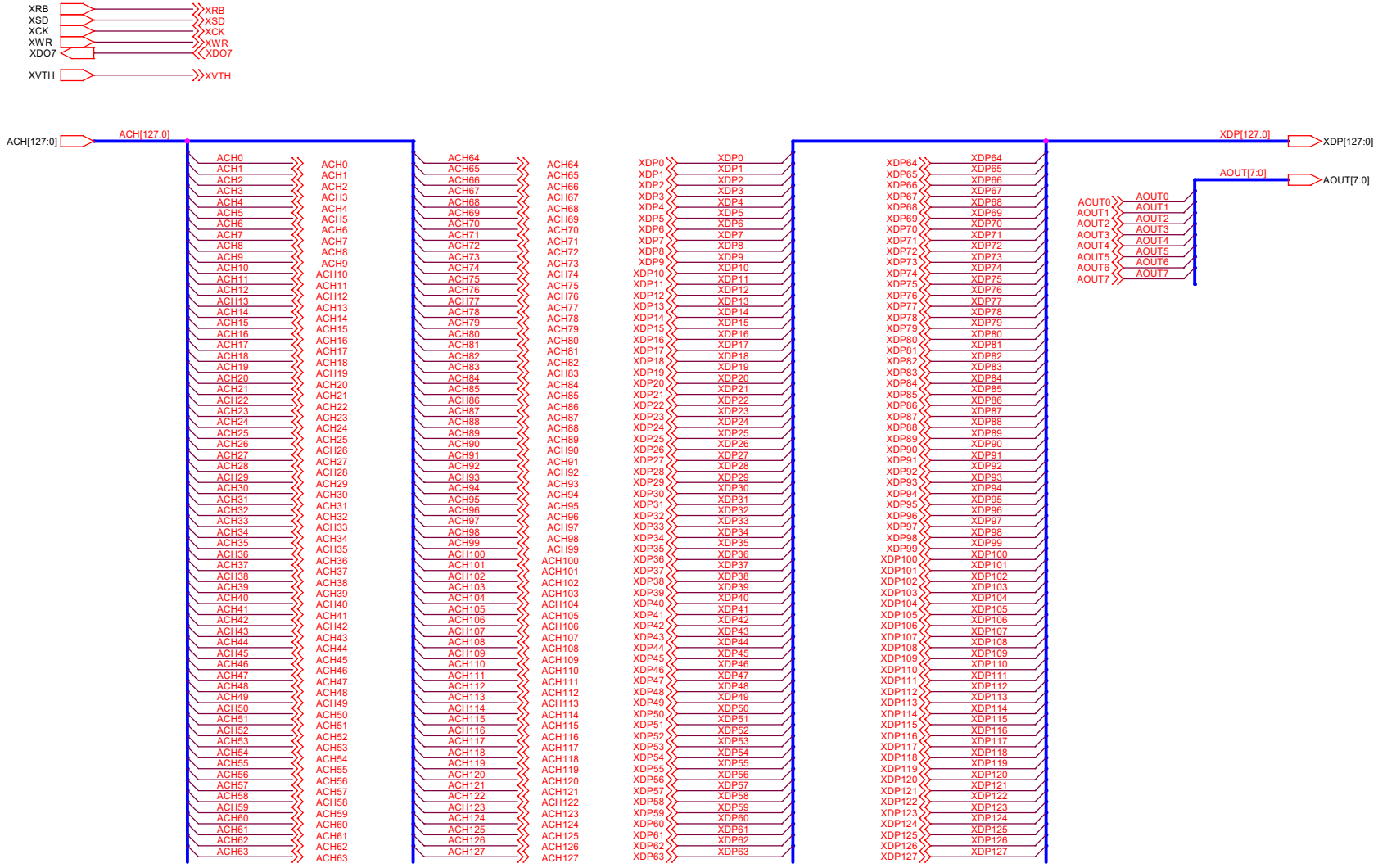
FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.
Title: INPUT_128		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 3 of 25	



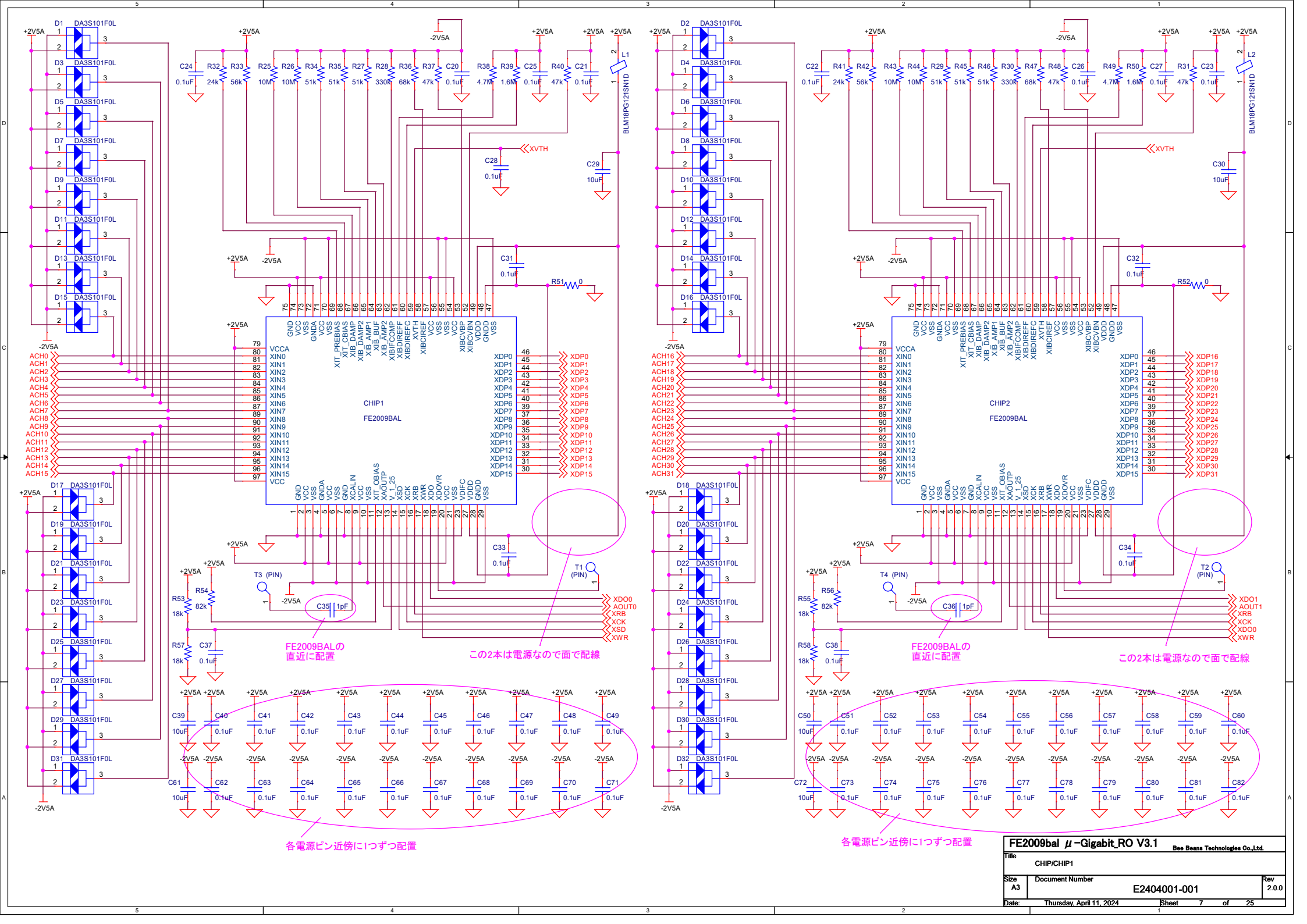
FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.	
Title: CHIP_DAC			
Size: A3	Document Number: E2404001-001		Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 4 of 25		



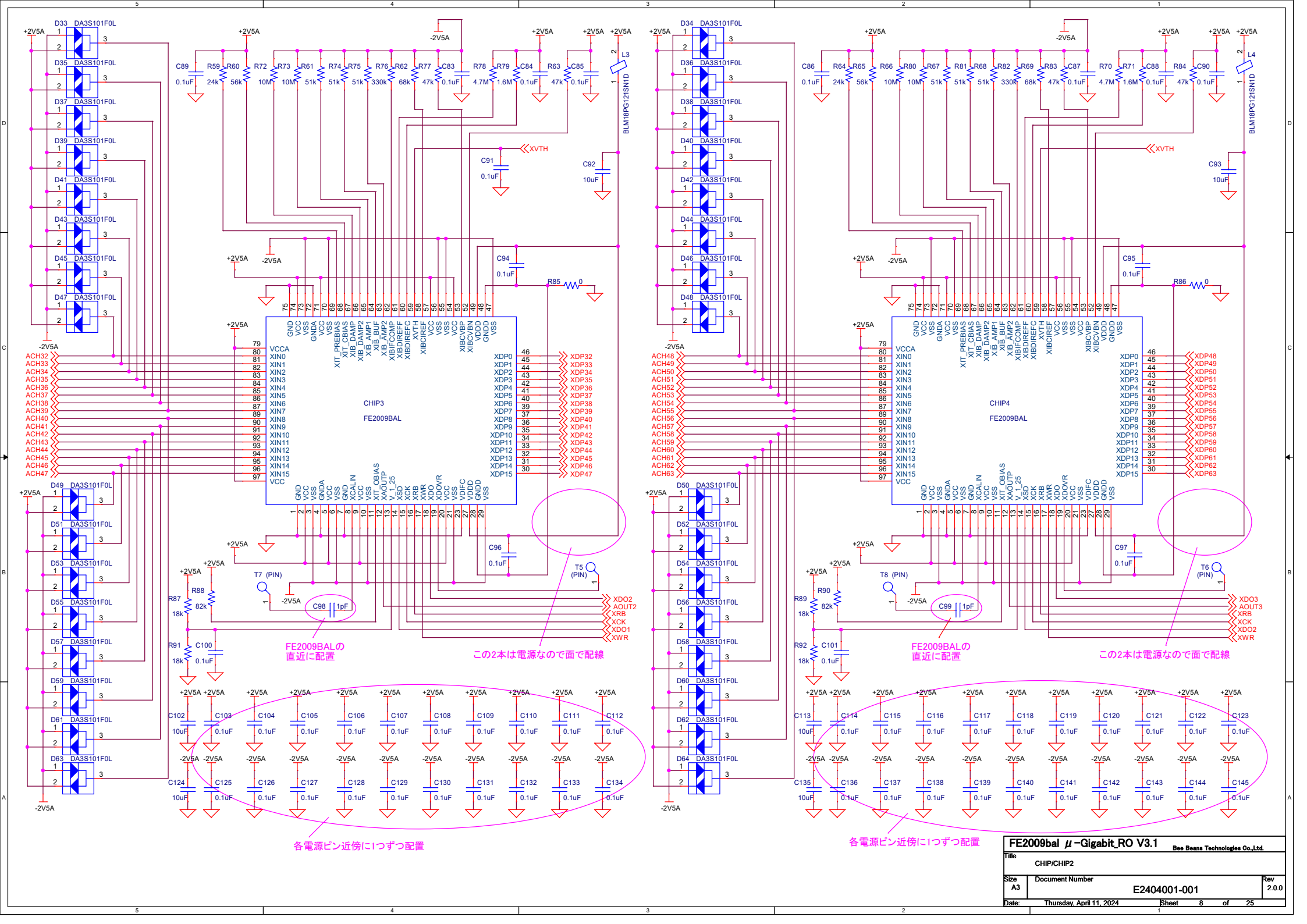
FE2009bal μ-Gigabit_RO V3.1	Bee Beans Technologies Co.,Ltd.	
Title: ADC_BIAS_DAC		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 5 of 25	



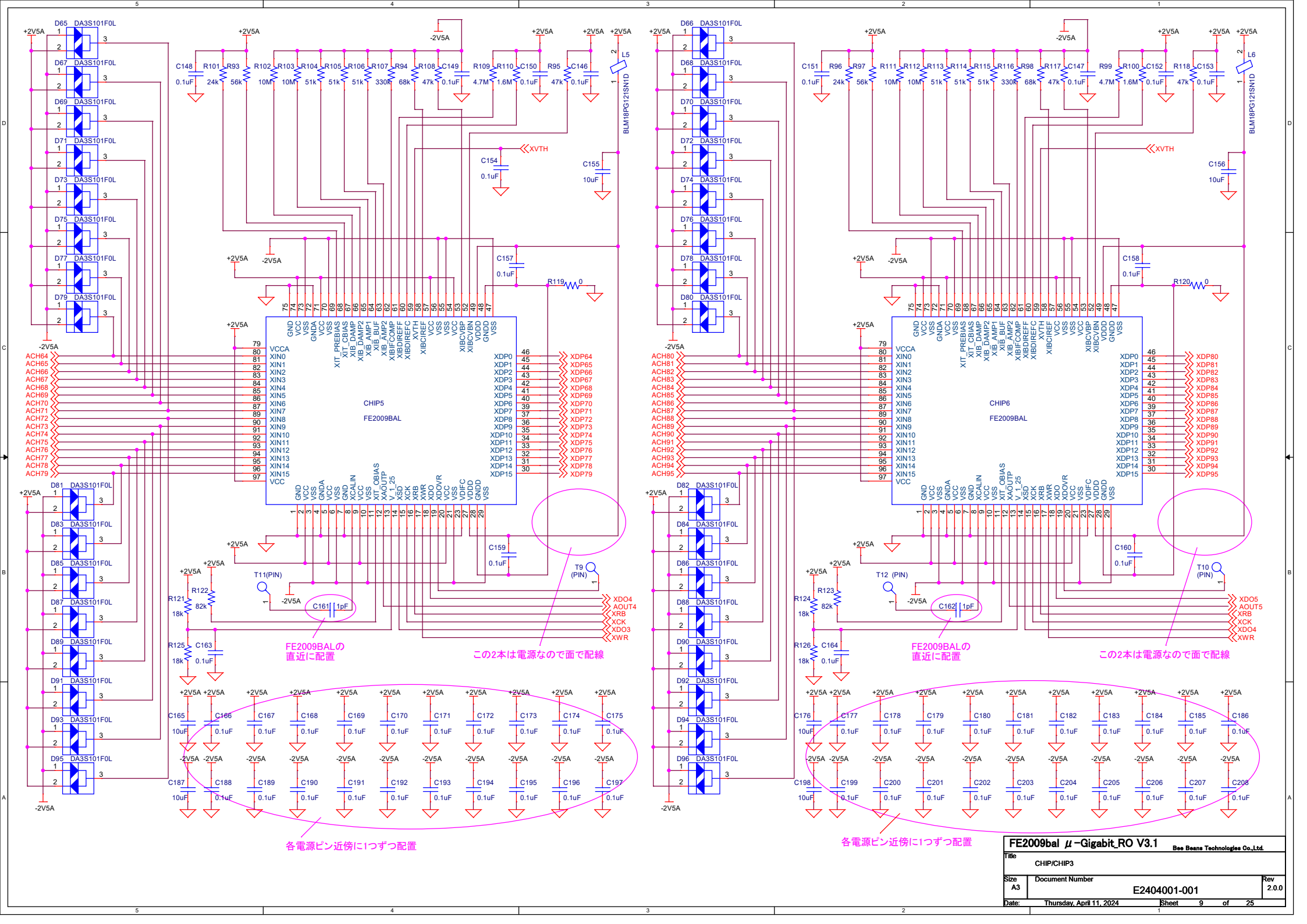
FE2009bal μ-Gigabit_RO V3.1	Bee Beans Technologies Co.,Ltd.	
Title: CHIP/CHIP0		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 6 of 25	



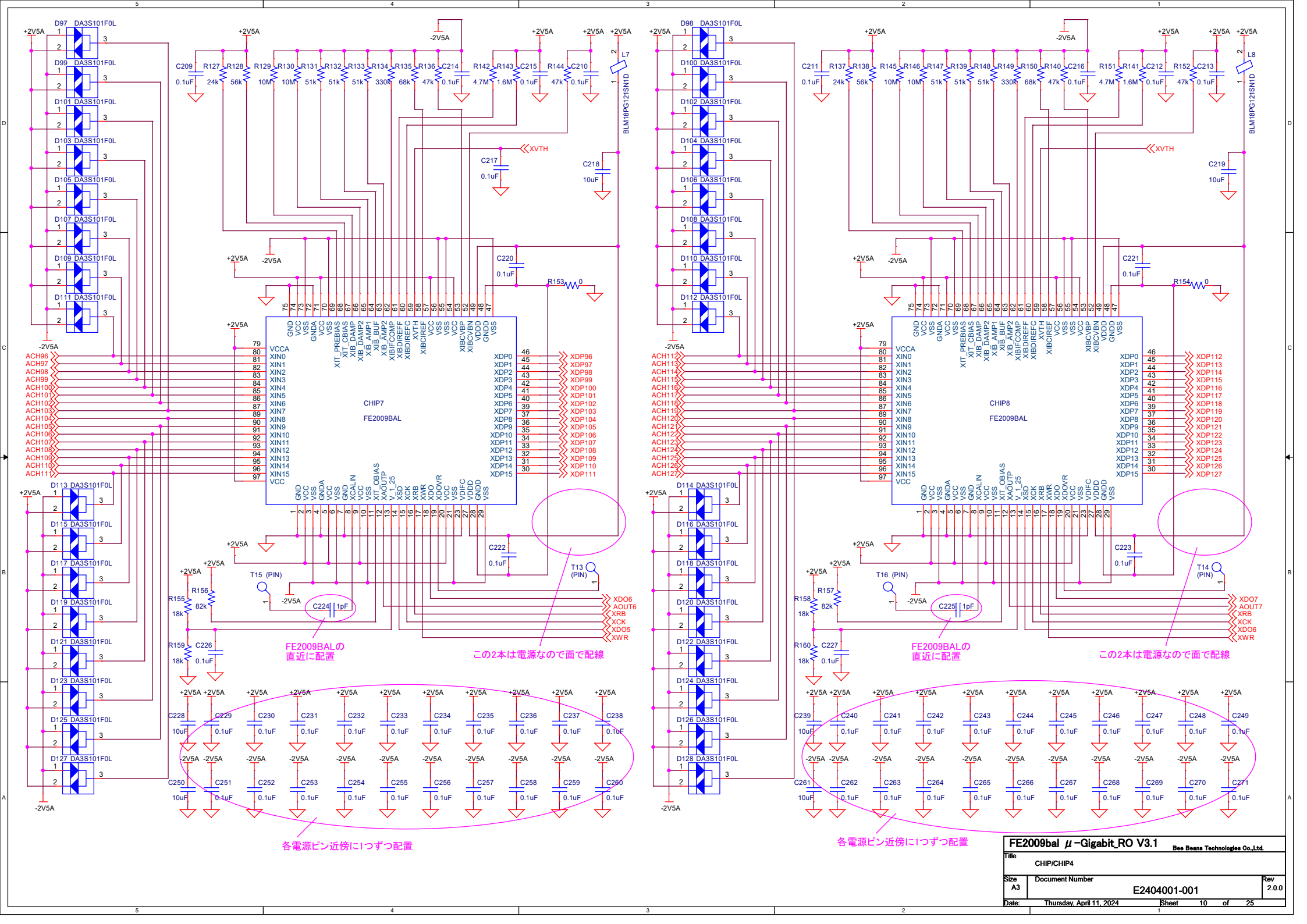
FE2009bal μ-Gigabit_RO V3.1	Bee Beans Technologies Co.,Ltd.	
Title: CHIP/CHIP1		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 7 of 25	



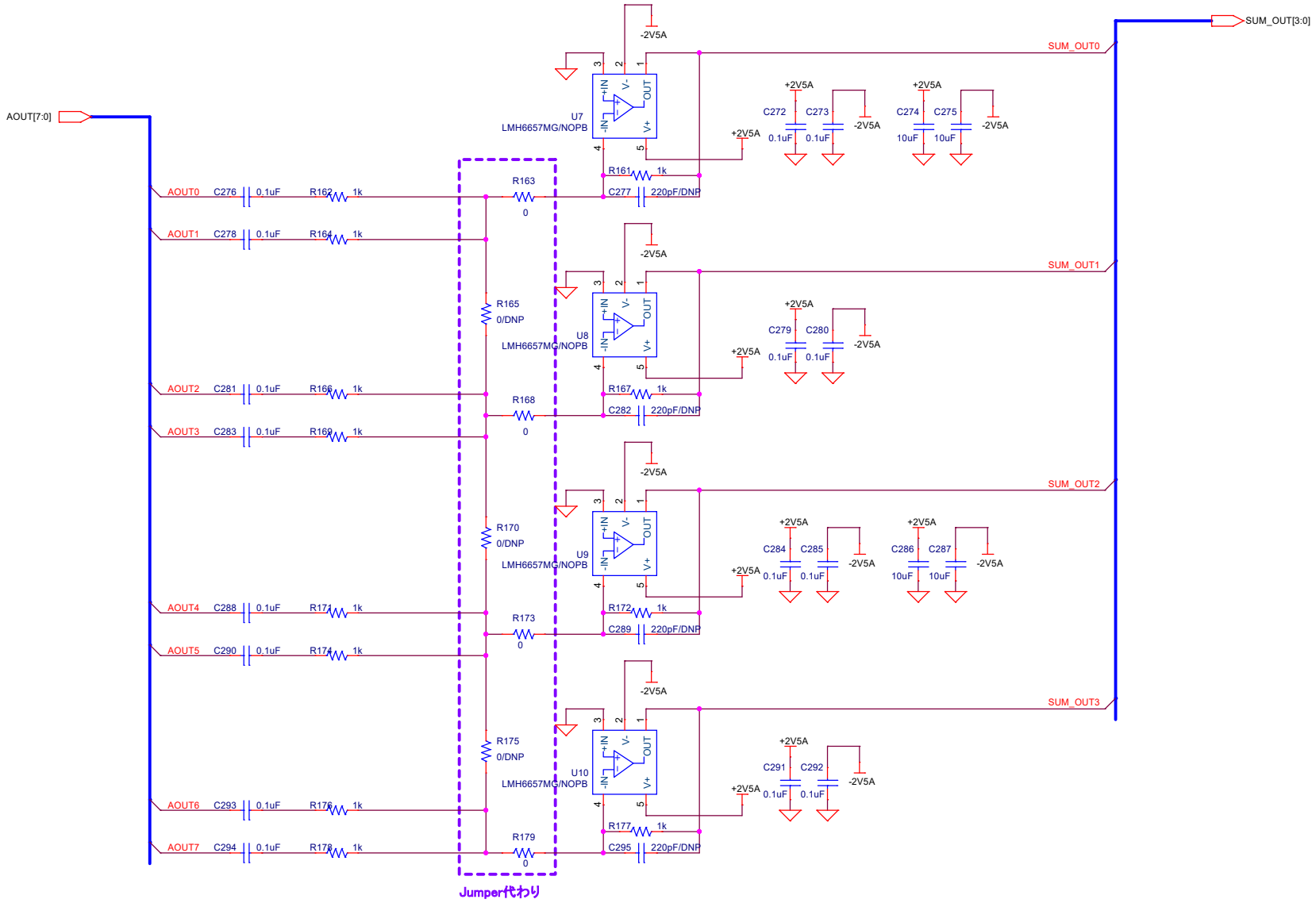
FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.
Title: CHIP/CHIP2		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 8 of 25	



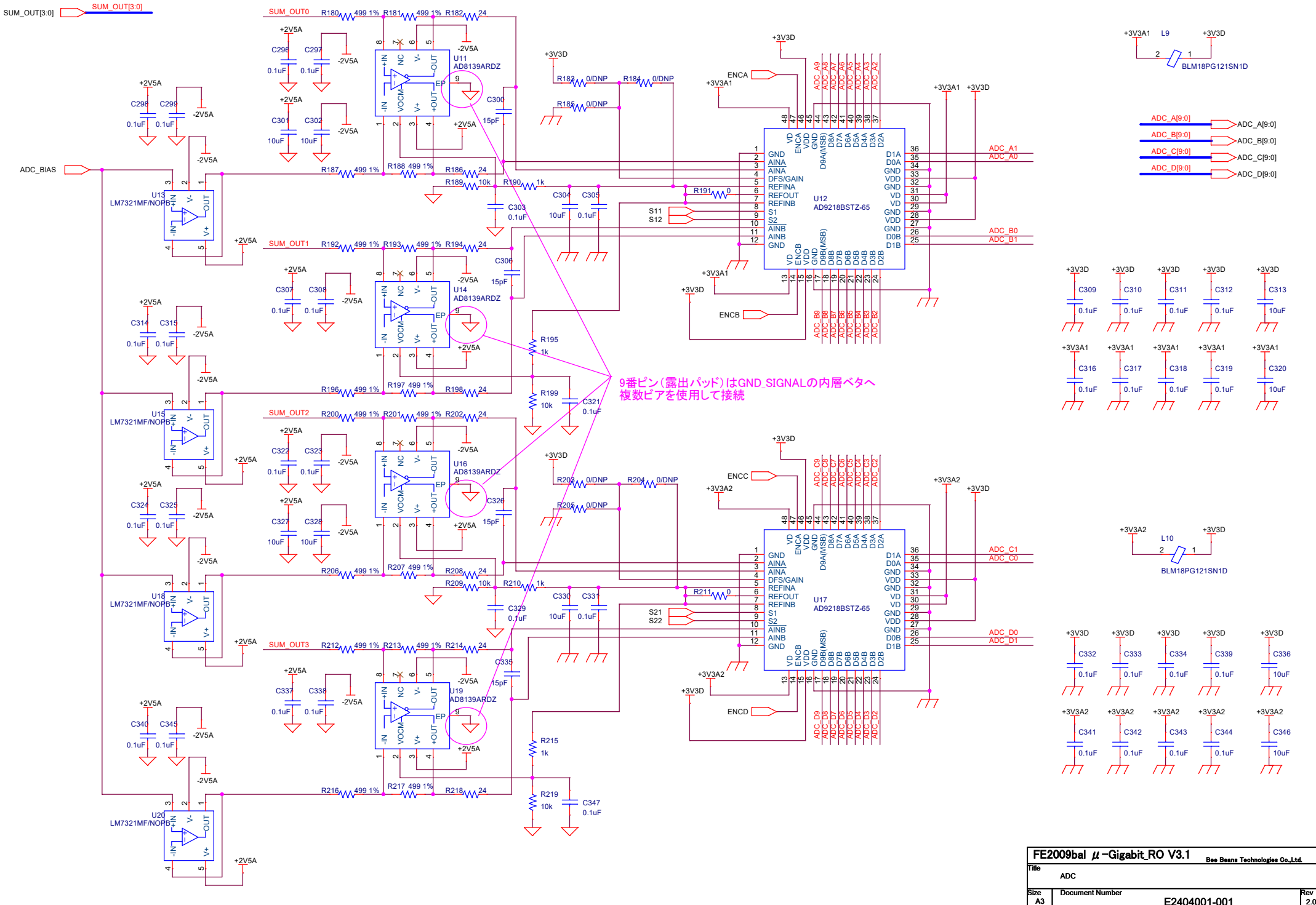
FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.
Title: CHIP/CHIP3		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
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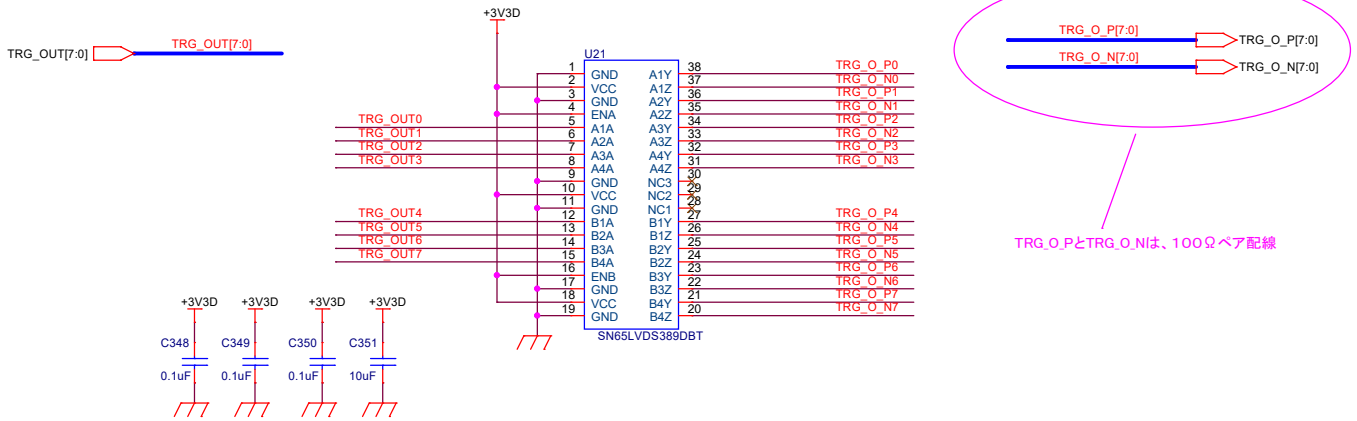
FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.	
Title: CHIP/CHIP4			
Size: A3	Document Number: E2404001-001		Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 10 of 25		

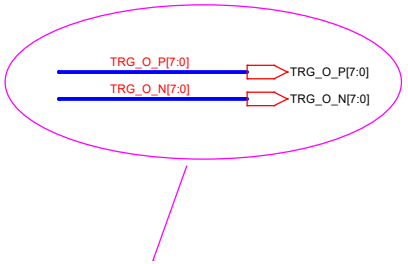


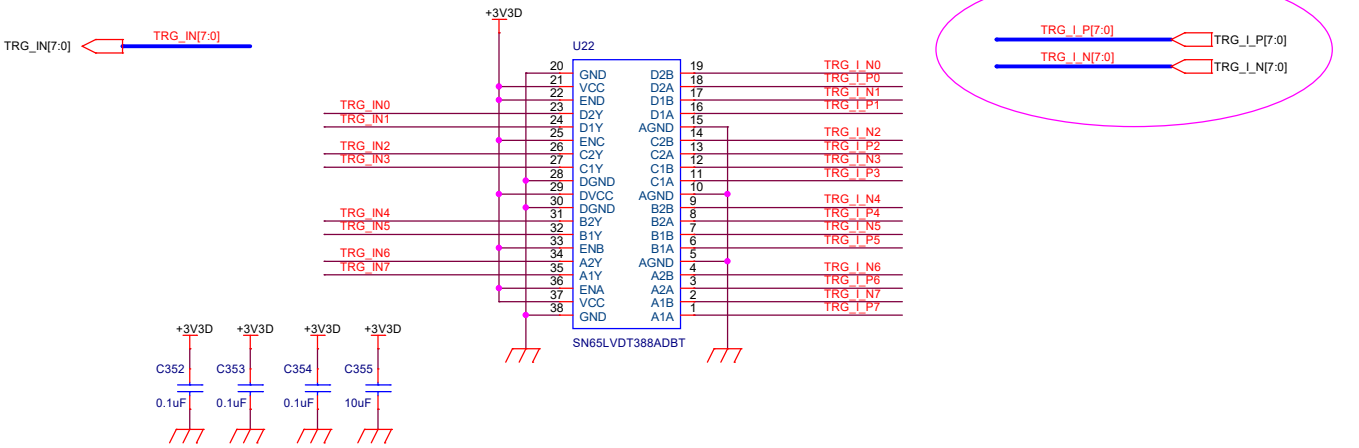
FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.
Title: ASUM		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 11 of 25	

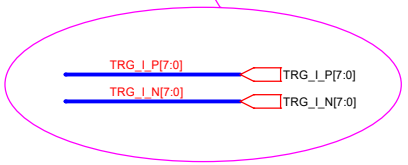


FE2009bal μ-Gigabit_RO V3.1	Bee Beans Technologies Co.,Ltd.	
Title: ADC		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 12 of 25	





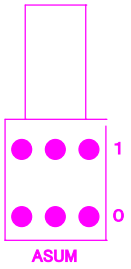




FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.
Title: LVDS		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
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【ASUM0～ASUM3のシルクイメージ】

コネクタ全体にASUMを表示し、ピンの列に番号を表示



SUM_OUT[3:0]

SUM_OUT0 — R220 49.9 1%/1005

SUM_OUT1 — R221 49.9 1%/1005

C356 1000pF, C357 1000pF

J1 (UP / LOW) — EPY.00.250.NTN

ASUM_0, ASUM_1

SUM_OUT2 — R222 49.9 1%/1005

SUM_OUT3 — R223 49.9 1%/1005

C358 1000pF, C359 1000pF

J2 (UP / LOW) — EPY.00.250.NTN

ASUM_2, ASUM_3

LED[3:0]

LED0 — R224 680

LED1 — R225 680

LED2 — R226 680

LED3 — R227 680

D129 SML-A12MTT86, D130 SML-A12MTT86, D131 SML-A12MTT86, D132 SML-A12UTT86

基板端方向から見えるよう
基板裏面に配置

MODE1, MODE2, RUN, HIT

TRG_I_P[7:0]

TRG_I_N[7:0]

TRG_O_P[7:0]

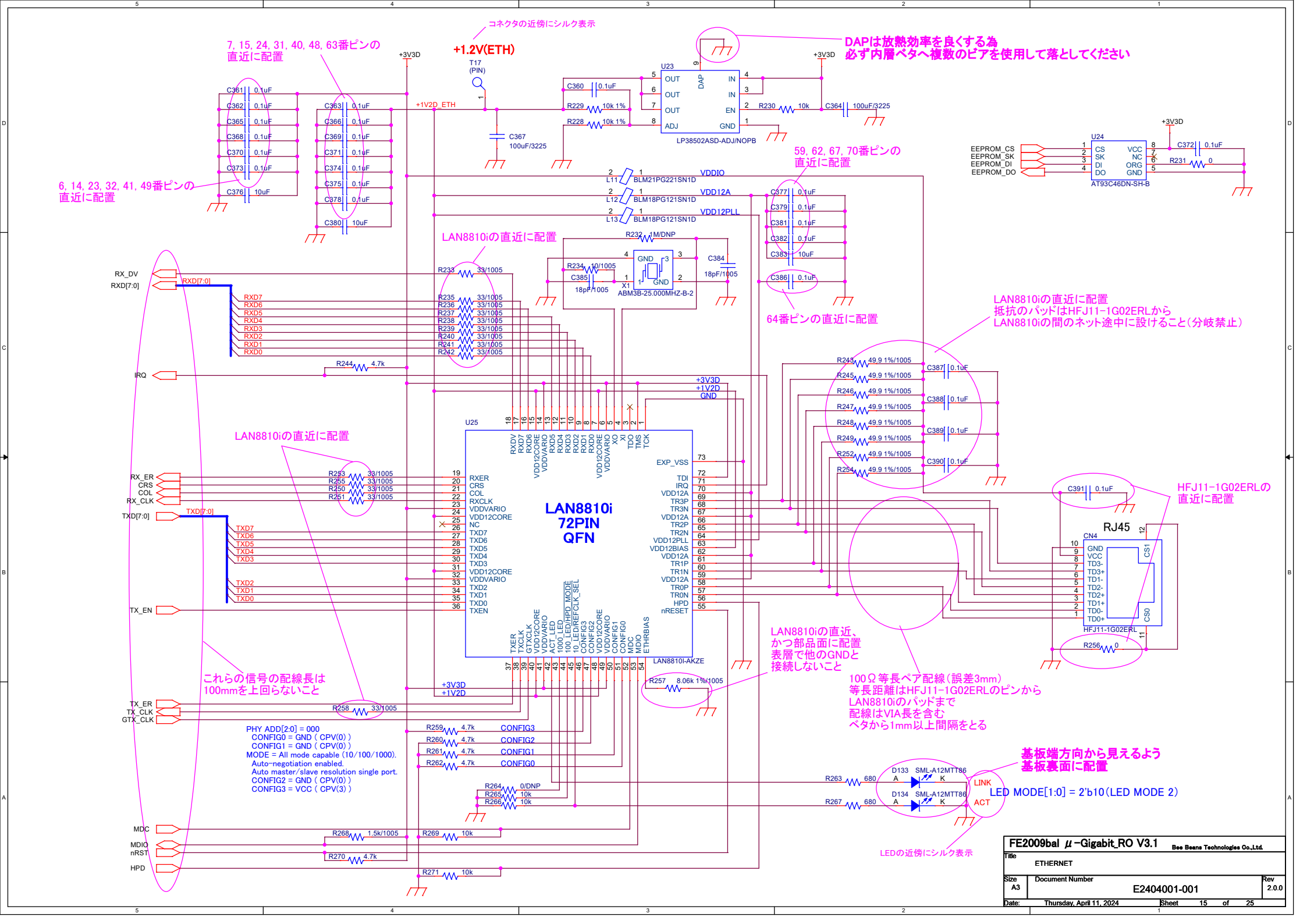
TRG_O_N[7:0]

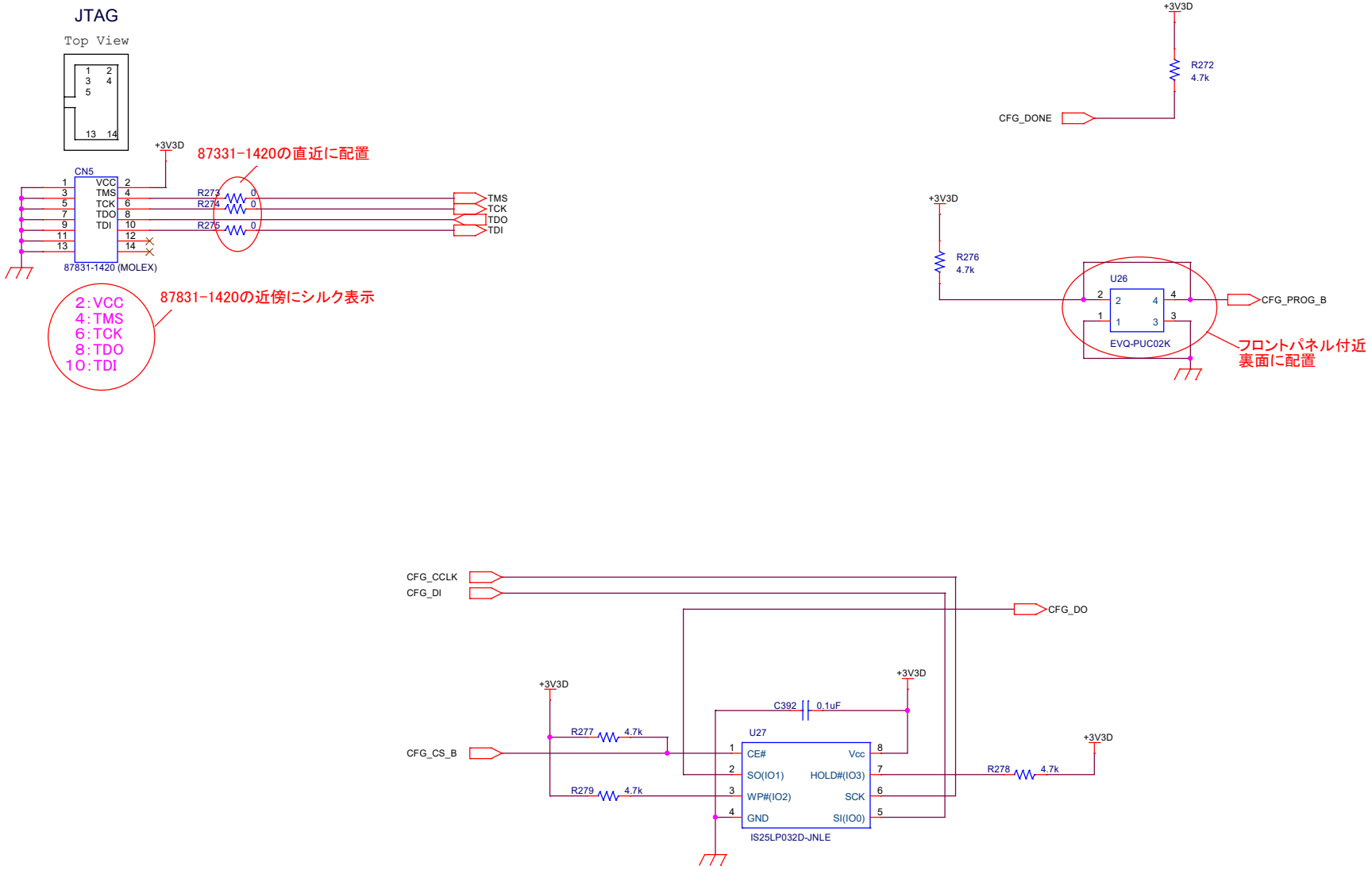
CN3

TRG_O_P0	1	2	TRG_O_N0
TRG_O_P1	3	4	TRG_O_N1
TRG_O_P2	5	6	TRG_O_N2
TRG_O_P3	7	8	TRG_O_N3
TRG_O_P4	9	10	TRG_O_N4
TRG_O_P5	11	12	TRG_O_N5
TRG_O_P6	13	14	TRG_O_N6
TRG_O_P7	15	16	TRG_O_N7
TRG_I_P0	17	18	TRG_I_N0
TRG_I_P1	19	20	TRG_I_N1
TRG_I_P2	21	22	TRG_I_N2
TRG_I_P3	23	24	TRG_I_N3
TRG_I_P4	25	26	TRG_I_N4
TRG_I_P5	27	28	TRG_I_N5
TRG_I_P6	29	30	TRG_I_N6
TRG_I_P7	31	32	TRG_I_N7
	33	34	

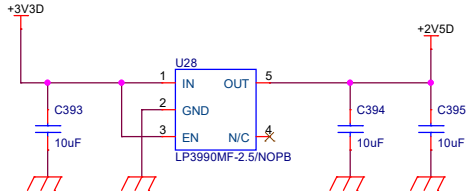
HIF3BA-34PA-2.54DS(71)

FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.	
Title: PANEL			
Size: A3	Document Number: E2404001-001		Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 14 of 25		

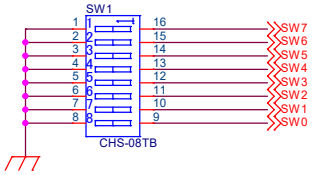

コネクタの近傍にシルク表示
+1.2V(ETH)
7, 15, 24, 31, 40, 48, 63番ピンの
直近に配置
6, 14, 23, 32, 41, 49番ピンの
直近に配置
DAPは放熱効率を良くする為
必ず内層ベタへ複数のビアを使用して落としてください
LP38502ASD-ADJ/NOPB
59, 62, 67, 70番ピンの
直近に配置
64番ピンの直近に配置
LAN8810iの直近に配置
LAN8810i
72PIN
QFN
LAN8810I-AKZE
LAN8810iの直近に配置
抵抗のパッドはHFJ11-1G02ERLから
LAN8810iの間のネット途中に設けること(分岐禁止)
HFJ11-1G02ERLの
直近に配置
RJ45
HFJ11-1G02ERL
LAN8810iの直近、
かつ部品面に配置
表層で他のGNDと
接続しないこと
100Ω等長ペア配線(誤差3mm)
等長距離はHFJ11-1G02ERLのピンから
LAN8810iのパッドまで
配線はVIA長を含む
ベタから1mm以上間隔をとる
これらの信号の配線長は
100mmを上回らないこと
PHY ADD[2:0] = 000
CONFIG0 = GND (CPV(0))
CONFIG1 = GND (CPV(0))
MODE = All mode capable (10/100/1000).
Auto-negotiation enabled.
Auto master/slave resolution single port.
CONFIG2 = GND (CPV(0))
CONFIG3 = VCC (CPV(3))
基板端方向から見えるよう
基板裏面に配置
LED MODE[1:0] = 2'b10(LED MODE 2)
LEDの近傍にシルク表示
AT93C46DN-SH-B
BLM21PG221SN1D
BLM18PG121SN1D
ABM3B-25.000MHZ-B-2
FE2009bal μ-Gigabit_RO V3.1
Bee Beans Technologies Co.,Ltd.
ETHERNET
E2404001-001
Thursday, April 11, 2024



FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.
Title: FPGA_PROM		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 16 of 25	

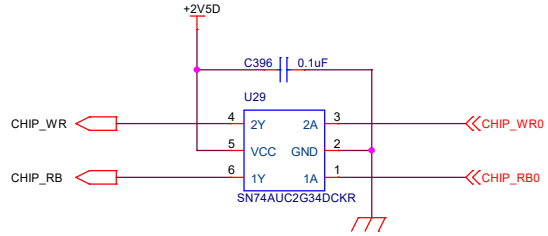


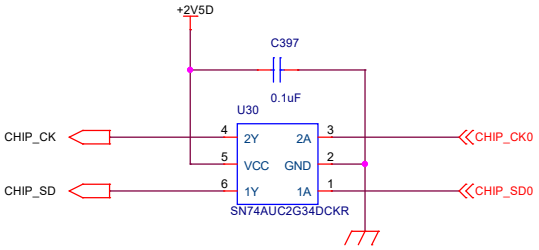
BANK_East(BANK 15)



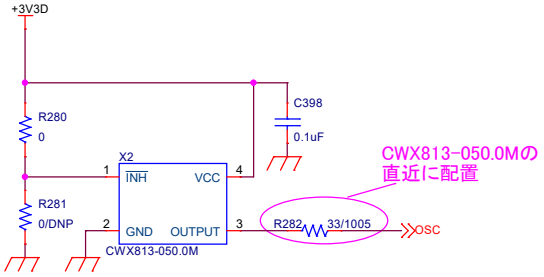
CHS-08TBの配置は任意

BANK_South(BANK 33)

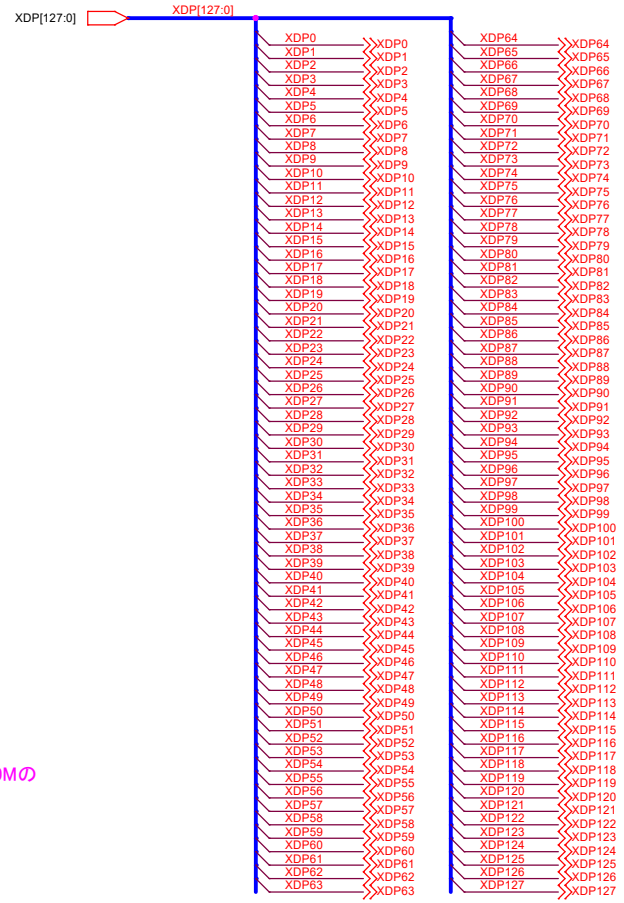




BANK_East(BANK 14)



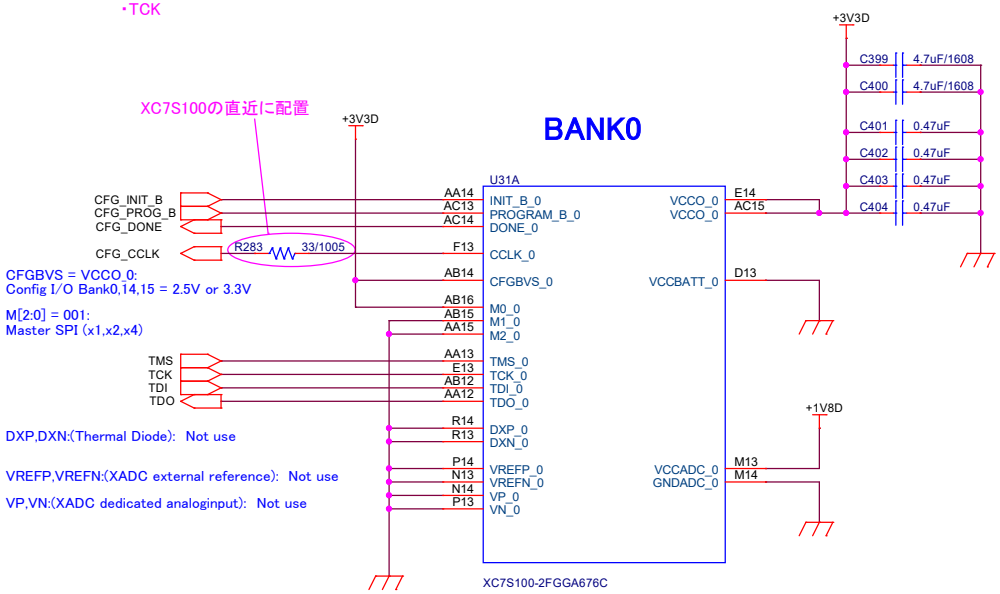
BANK_North(BANK 36)
BANK_West(BANK 35,34)
BANK_South(BANK 33)



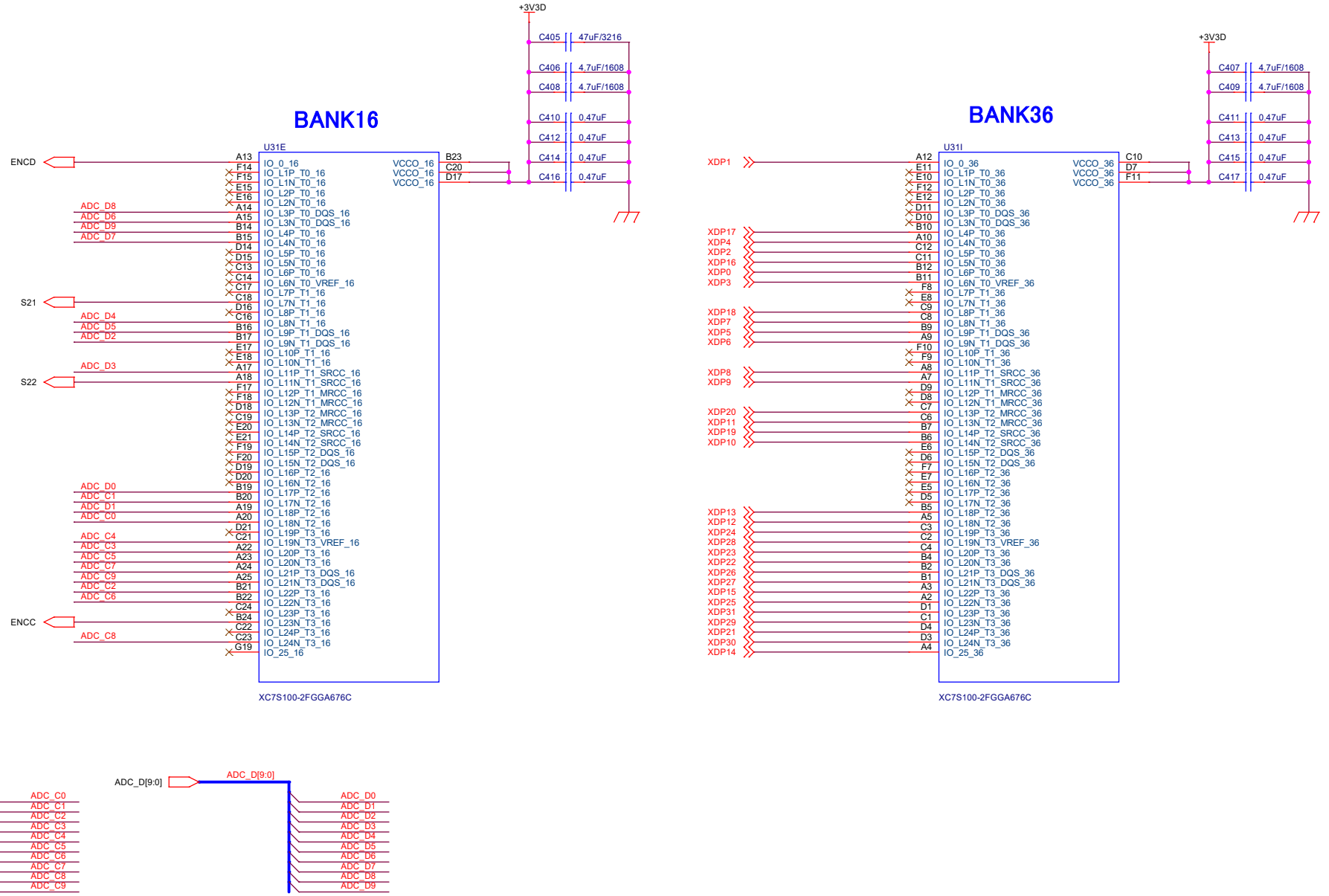
FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.
Title: FPGA/FPGA_MISC		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 17 of 25	

【優先ネット】
以下のネットは、優先配線してください。

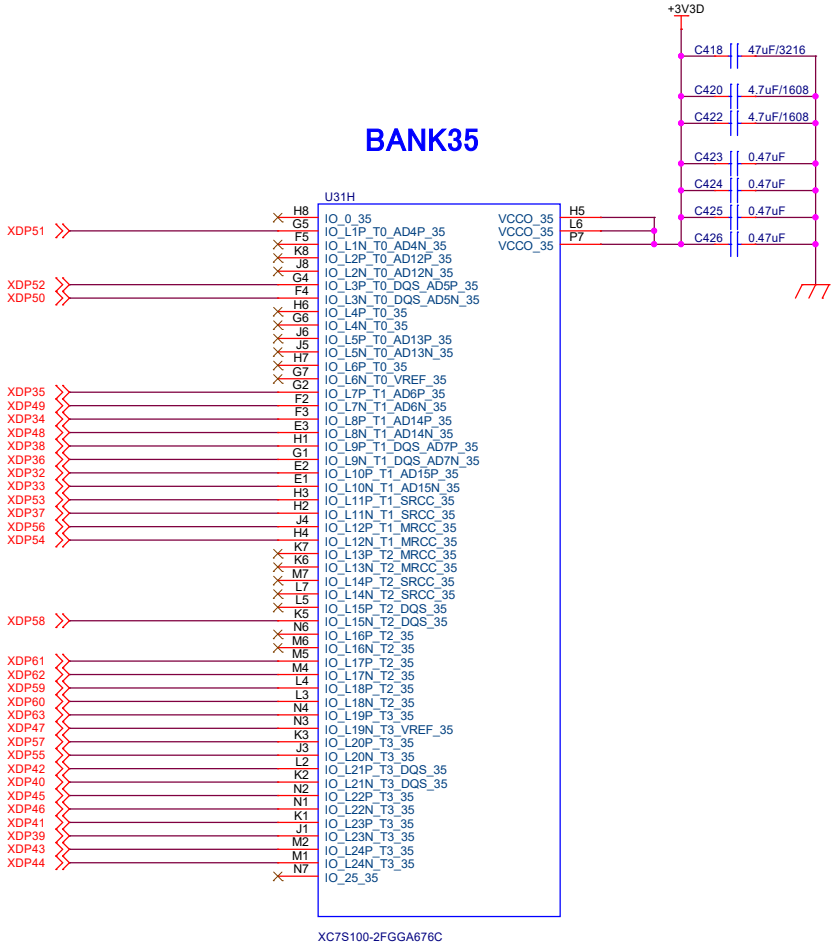
・CFG_CCLK
・TCK

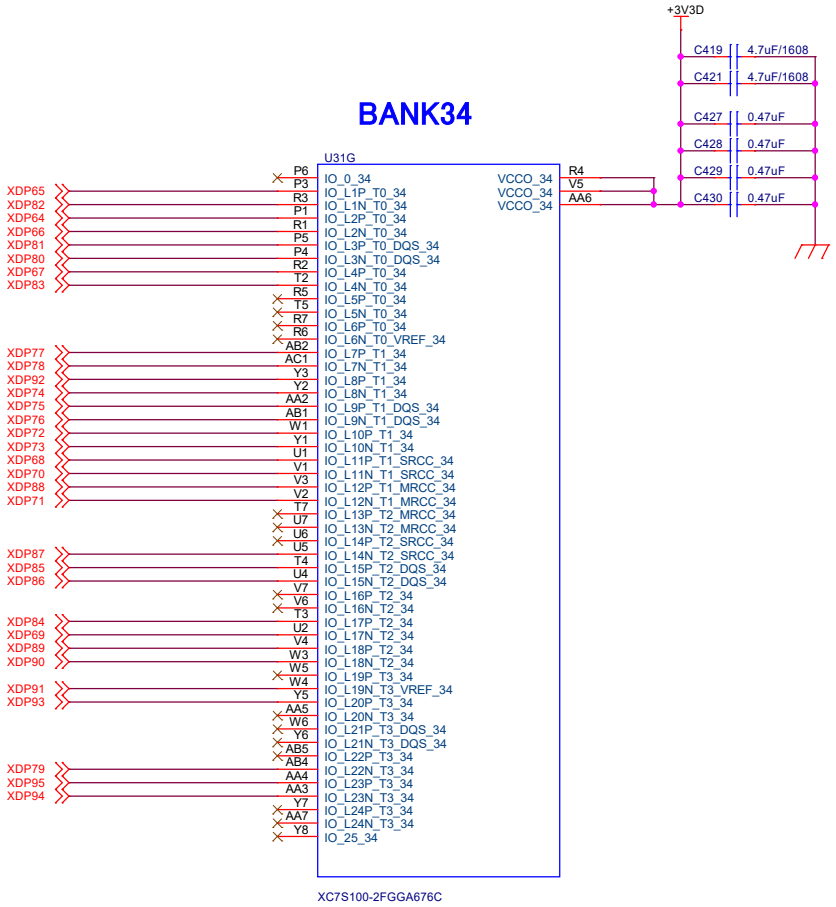


FE2009bal μ-Gigabit_RO V3.1	Bee Beans Technologies Co.,Ltd.	
Title: FPGA/FPGA_BANK0		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 18 of 25	

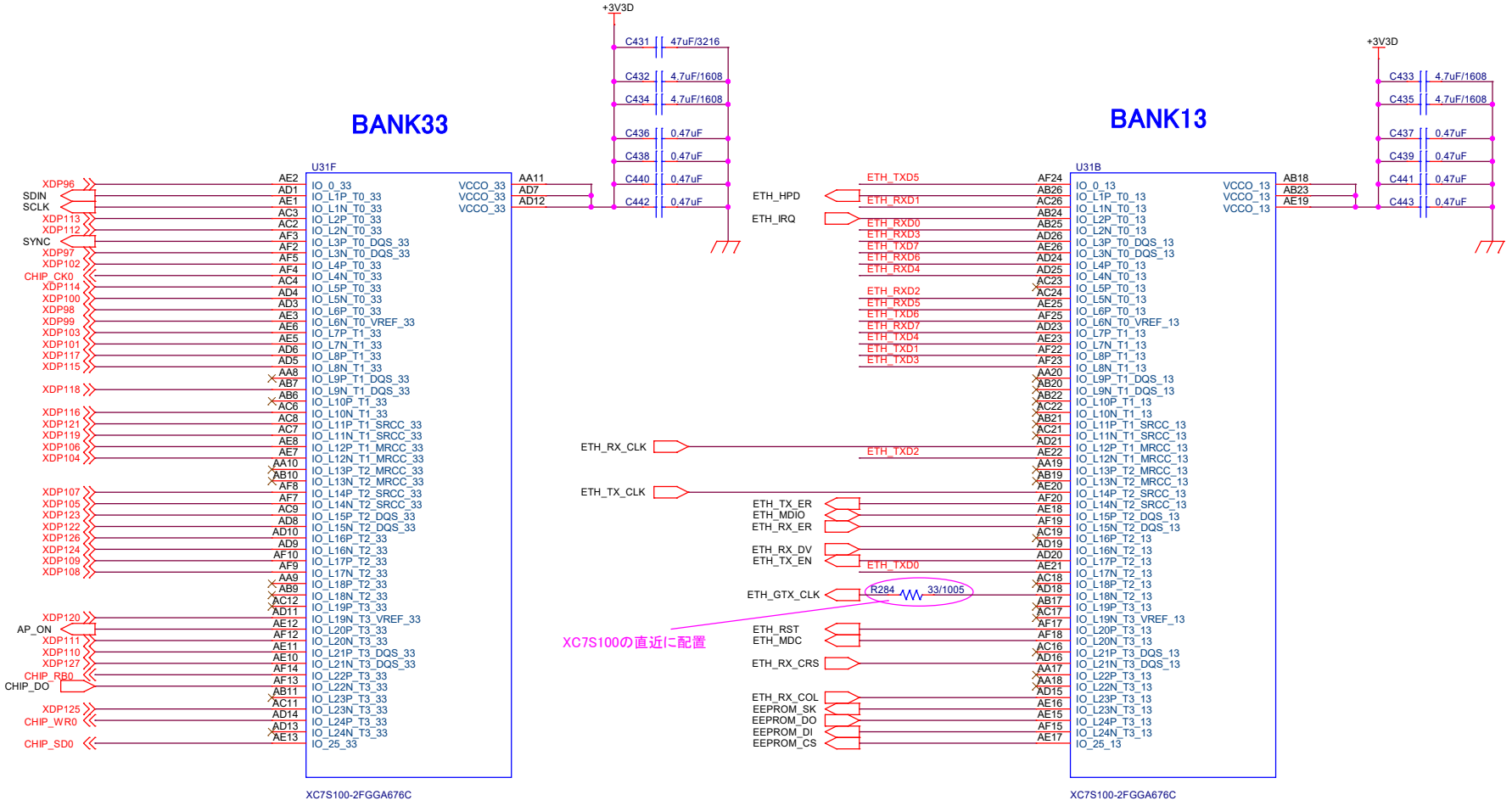


FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.
Title: FPGA/FPGA_NorthBANK		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
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FE2009bal μ-Gigabit_RO V3.1		Bee Beans Technologies Co.,Ltd.
Title: FPGA/FPGA_WestBANK		
Size: A3	Document Number: E2404001-001	Rev: 2.0.0
Date: Thursday, April 11, 2024	Sheet 20 of 25	



【優先配線ネット】
以下のネットは優先配置・優先配線してください

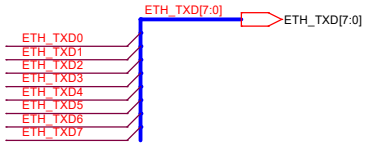
- ETH_RX_CLK
- ETH_TX_CLK
- ETH_GTX_CLK
- ETH_TXD[7:0]
- ETH_TX_EN
- ETH_TX_ER
- ETH_RXD[7:0]
- ETH_RX_DV
- ETH_RX_ER
- ETH_RX_COL
- ETH_RX_CRS

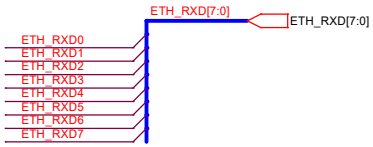
【等長配線A】
以下のネットは等長（誤差±10mm）としてください

- ETH_TX_CLK（基準ネット）
- ETH_GTX_CLK（基準ネット）
- ETH_TXD[7:0]
- ETH_TX_EN
- ETH_TX_ER

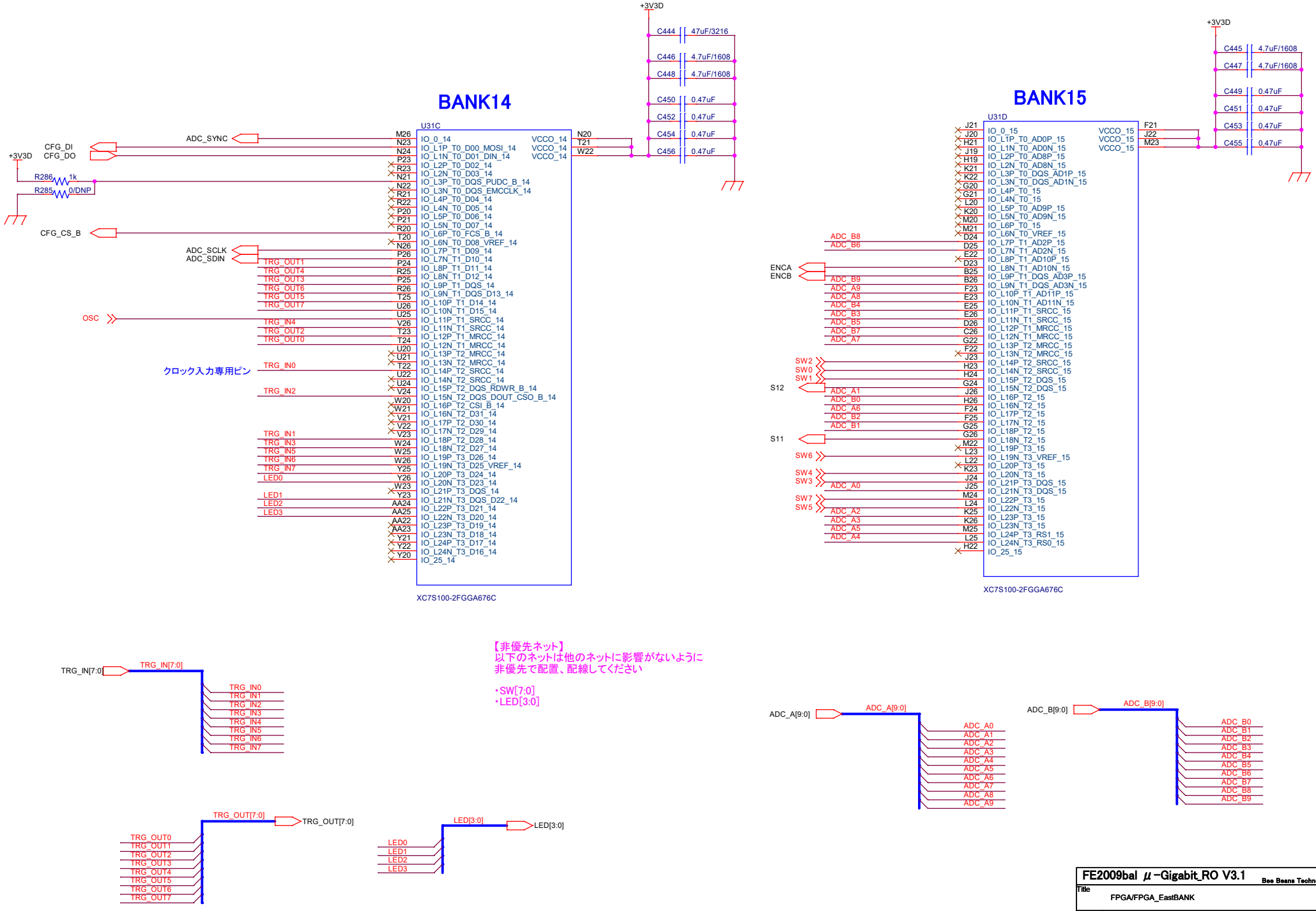
【等長配線B】
以下のネットは等長（誤差±10mm）としてください

- ETH_RX_CLK（基準ネット）
- ETH_RXD[7:0]
- ETH_RX_DV
- ETH_RX_ER
- ETH_RX_COL
- ETH_RX_CRS



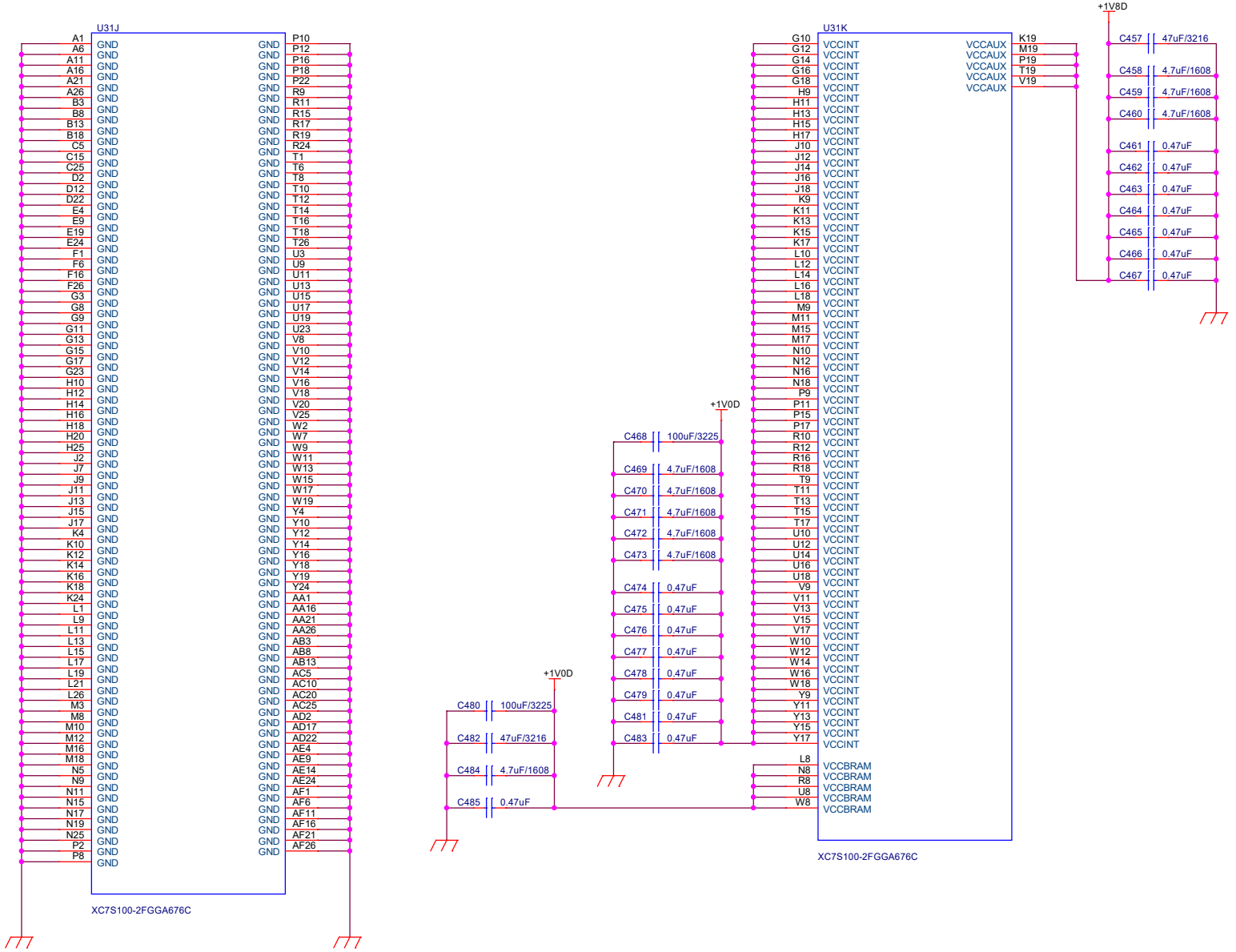


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POWER



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Vo=3.3V±0.165V

+3.3V

DGND

CN6

CN7

TH-1.6-5.0-M3

F1

0603SFF500F/32-2

L14

NFM18PS105R0J3D

+3V3P

コネクタの近傍にシルク表示

テストVIAの近傍にシルク表示

D +1.0V

D +1.8V

D +3.3V

T18 (PIN)

T19 (PIN)

T20 (PIN)

+1V0D

+1V8D

+3V3D

テストポイントの近傍にシルク表示

D GND

D GND

TP1 JP-4

TP2 JP-4

部品資料の参考パタン参照

U32

TPS22954DSQR

IN

IN

BIAS

EN

GND

OUT

OUT

SNS

PG

CT

EP

C487 22uF/3216

C488 4700pF

C486 100uF/3225

R287 5.6k 1%

R288 4.7k

R289 1.3k 1%

R290 3.9k 1%

R291 1.3k 1%

R292 10k 1%

Ct=4700pF
Tr=4246us

PG
VIH=2.4V to 3.1V
VIL=2.1V to 2.7V

UVLO
VIH=2.5V to 3.0V
VIL=2.1V to 2.6V

CFG_INIT_B

部品資料の参考パタン参照

U33

TPS7A9201DSKT

IN

IN

NR/SS

EN

SS_CTRL

OUT

OUT

FB

GND

PG

Thermal_Pad

C491 22uF/3216

C492 0.1uF

C489 22uF/3216

C490 0.01uF

R293 2.0k 1%

R294 1.6k 1%

R295 10k

Vo=1.762V to 1.838V

SS_CTRL=VIN
Ct=0.1uF(;20%/+80%)
Tr:0.3 to 2.1ms

部品資料の参考パタン参照
Thermal_Padは、放熱のため半田面にもGNDプレーンを配置してください

U34

TPS7A9201DSKT

C495 22uF/3216

C496 0.1uF

C493 22uF/3216

C494 0.01uF

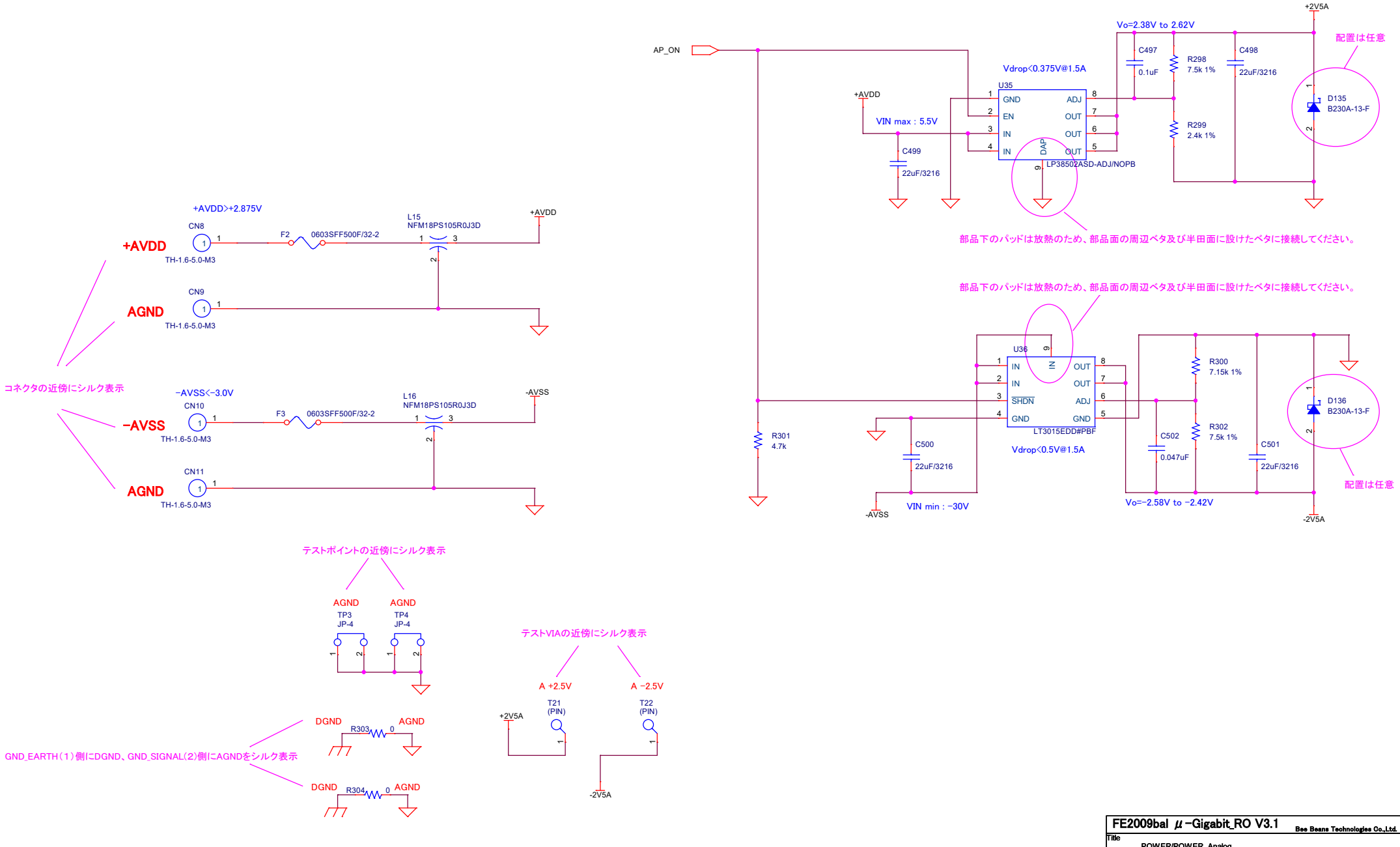
R296 2.7k 1%

R297 10k 1%

Vo=1.002V to 1.031V

SS_CTRL=VIN
Ct=0.1uF(;20%/+80%)
Tr:0.3 to 2.1ms

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